Tree Chatbot BETA

+ New chat

WORKSPACE

Product research 6

PRODUCT RESEARCH

Cut ⌘X

Copy ⌘C

Paste ⌘V

Rename ↵

Delete ⌫

RECENT

API rate limits

Weekly planning

Untitled chat

Your keys. Your conversations.

Saved locally in this browser.

JD Personal workspace

Local · no account needed

PROVIDER Gemini

MODEL gemini-2.5-pro

03

key saved ••••1234 Change Remove

All changes saved locally

CHOOSE A PROVIDER

Gemini ✓

Claude Add key

OpenAI Add key

Grok Add key

Each provider uses its own API key.

Search architecture

Gemini · Root chat

How should we design search for our docs? Compare a few approaches.

ASSISTANT

Start with hybrid search.
Combine exact keyword matches with semantic retrieval.

1. **Vector search** for intent
2. **Reranking** for relevance

Ask a follow-up...

Branch 2 children Delete

Delete child chats before deleting this chat.

Embeddings

Vector search deep dive

Claude · Branched from Search architecture

Let's explore embeddings. How do we measure similarity?

ASSISTANT

Use cosine similarity.
Embed each document and query into the same vector space.

$$\cos(\theta) = (a \cdot b) / (\|a\| \|b\|)$$

Ask a follow-up...

Branch Delete

Ranking

Ranking & evaluation

OpenAI · Branched from Search architecture

How do we combine keyword and vector results?

ASSISTANT

Reciprocal rank fusion (RRF)
Merge both ranked lists without normalizing their raw scores.

Start with k = 60, then evaluate on real queries using NDCG@10.

Ask a follow-up...

Branch Delete

Chat pasted into folder "Product research" Undo

100%

Scroll to zoom · Space + drag to pan

MINIMAP